**Printed Pages:02** **Sub Code: KAU 072**

**Paper Id: 231906** **Roll No.** | | | | | | | | | | |

**B.TECH**
**(SEM-VII) THEORY EXAMINATION 2022-2023**
# HYBRID VEHICLE PROPULSION

***Time: 3 Hours*** ***Total Marks: 100***

**Note:** Attempt all Sections. If require any missing data; then choose suitably.

## SECTION A

**1. Attempt *all* questions in brief. 2 x 10 = 20**

(a) What is the impact of modern drive-trains on energy supplies?
(b) Why an energy management control system is required in an HEV?
(c) Compare performance of EV and HEV.
(d) What are the supporting subsystems in EV and HEV?
(e) What are the different subsystems of electric drive train?
(f) What is the basic concept of electric traction?
(g) Differentiate between complex hybrid and series hybrid
(h) Give the advantages and disadvantages of fuel cells.
(i) What are the features of motor used in EV?
(j) List the four example of rule based strategies that can be applied to energy management in EV?

## SECTION B

**2. Attempt any *three* of the following: 10 x 3 = 30**

(a) Draw a general lay out of an EV and discuss the transmission characteristics.
(b) With the help of a neat block diagram explain different subsystems of electric drive train.
(c) Describe about direct rotor flux orientation schemes of induction motor drives used in hybrid electric vehicles.
(d) How is it possible to use fly wheel technologies as an energy storage devicein electric vehicles?
(e) Discuss in brief the different implementation issues of energy management strategies.

## SECTION C

**3. Attempt any *one* part of the following: 10 x 1 = 10**

(a) What are the social and environmental impacts of Hybrid and Electric Vehicles?
(b) Describe the vehicle power source characterization and mathematical models to describe vehicle performance.

**4. Attempt any *one* part of the following: 10 x 1 = 10**

(a) Explain the Series - Parallel Configurations of Hybrid Drive Train with neat diagram.

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(b) Explain the different power flow control modes of a typical parallel hybrid system with help of block diagram.

**5. Attempt any *one* part of the following: 10 x 1 = 10**

(a) Explain the Multi-quadrant control of DC motors used in hybrid electric vehicles.

(b) Explain the extension of speed technology of PMMC motor drives in hybrid electric vehicles.

**6. Attempt any *one* part of the following: 10 x 1 = 10**

(a) Describe the basic principle of super capacitors based energy storage system in hybrid electric vehicles.

(b) Explain the sizing procedures of energy storage systems for electric hybrid vehicles.

**7. Attempt any *one* part of the following: 10 x 1 = 10**

(a) Explain the different categories of energy management strategies in electric vehicles and hybrid electric vehicles.

(b) Explain the following briefly

i) Sizing the propulsion motor

ii) sizing the power electronics

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